Elastic–Causal Storage and Delayed Release of Energy in SN 2023zkd

A CET validation test with ATLAS+ZTF photometry

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Abstract

We present a data-driven validation of the Causal Elastic Theory (CET) through an extended photometric investigation of the Type Ibn supernova SN 2023zkd, a rare event displaying a distinctive two-peaked luminosity evolution. Using a combination of TNS discovery photometry, ATLAS multi-band coverage, and ZTF forced-photometry light curves, we reconstruct the bolometric emission and color evolution across more than one year. The resulting light curve reveals a complex bolometric morphology, including a pronounced secondary maximum and non-trivial color changes. By cross-matching with galaxy catalogs, we further characterize the explosion site as a low-density environment.

Within CET, the observed morphology is naturally explained by *elastic energy storage* in a dissipative, low-coupling medium, followed by a *delayed release* during relaxation. We formalize this scenario through an effective coupling $K_{\text{eff}}(t)$ and a time-dependent relaxation timescale $\tau(t)$, fitting the model directly to the bolometric curve. Our analysis shows that the fraction of energy radiated in the delayed channel is non-zero, providing empirical evidence for causal storage and release. Finally, we connect this finding to prior CET evidence in the *low-density regime* from Gaia binaries, introducing the concept of *residual effective coupling* as a unifying descriptor across different astrophysical systems.

1 Introduction

The photometric and spectroscopic investigation of Type Ibn supernovae has gained momentum in recent years, with particular attention to events showing complex light-curve morphology. Among them, SN 2023zkd ($z = 0.056$) represents a remarkable case, as presented in the comprehensive analysis by Gagliano et al. (2025). Their work established the presence of a distinctive double-peaked luminosity profile, raising important questions regarding the physical mechanisms driving such behavior.

Building upon this foundation, the present study aims to provide a complementary perspective. By extending the photometric dataset through the incorporation of ATLAS multi-band observations and ZTF forced-photometry, we construct a more continuous bolometric curve and analyze the associated color evolution. In addition, we perform an environmental characterization based on galaxy counts in the surrounding region, thereby placing the transient within its local density context.

The combination of extended light-curve coverage and environmental mapping allows us to revisit the interpretation of the double-peaked morphology. While circumstellar medium (CSM) interaction remains the leading explanation, our results suggest that additional processes—such as temporary energy retention in the system and delayed release—may also contribute to the observed luminosity evolution.

2 Objective

The goal of this study is to investigate the unusual double-peaked bolometric morphology of SN 2023zkd through extended photometric monitoring and environmental mapping. Starting from discovery photometry reported by TNS, we combine ATLAS and ZTF light curves to produce a robust bolometric reconstruction, along with a detailed analysis of color evolution and energy release. This approach allows us to examine not only the photometric signatures of the explosion but also the dynamical behavior of the progenitor environment.

In the course of this analysis, we find indications that part of the explosion energy was not radiated immediately, but instead stored in non-radiative channels before being released at later times. This possibility—consistent with the framework of Causal Elastic Theory (CET)—motivates the introduction of effective coupling and relaxation as physical tools to quantify delayed emission in low-density environments.

3 Environment Analysis

The surrounding environment of SN 2023zkd has been investigated from two complementary perspectives: (i) the host galaxy and circumstellar medium (CSM) properties, as presented in Gagliano et al. (2025), and (ii) the extragalactic density field in the vicinity of the event, obtained from our own cross-match with the Sloan Digital Sky Survey (SDSS). This dual approach allows us to combine the detailed host-level information provided by Gagliano et al. with a broader contextual view of the large-scale environment.

3.1 Host Galaxy and CSM (from Gagliano et al. 2025)

Gagliano et al. carried out a multi-wavelength characterization of the host galaxy using GALEX, Pan-STARRS, SDSS, DES, and 2MASS photometry, fitting the spectral energy distribution (SED) with the `Prospector` code. The results indicate a system of extremely low mass ($\log M_\star \approx 7.93\,M_\odot$), low star-formation rate (SFR $\approx 0.01\,M_\odot\,\mathrm{yr}^{-1}$), and subsolar metallicity. This places the host of SN 2023zkd in the category of dwarf, metal-poor galaxies, similar to the environments often associated with SLSNe I and Ibn events.

In terms of immediate surroundings, Gagliano et al. estimated the presence of $\sim$ 5–6 $M_\odot$ of circumstellar material interacting with the ejecta. The modeling suggests that this CSM was structured in at least two distinct episodes of mass loss, one occurring roughly 3–4 years before the explosion and another about 1–2 years before. Within the canonical framework, this episodic mass ejection accounts for the unusual double-peaked light curve observed in SN 2023zkd. The host-level analysis in their study, however, is restricted to the stellar population and CSM interaction, without extending to the extragalactic density field.

3.2 Extragalactic Density Field (this work)

To complement the host-level analysis, we constructed a galaxy cone around the position of SN 2023zkd and cross-matched it with the SDSS spectroscopic and photometric catalogs. The resulting sample provides the number counts, magnitudes, and redshifts of neighboring galaxies within a projected radius of 1.500 00 Mpc (comoving).

Our analysis shows that SN 2023zkd lies in a markedly underdense region: the immediate environment contains only a sparse population of galaxies, with no evidence of massive clusters or rich filamentary substructure. The median galaxy density is significantly below the SDSS field average, confirming that the event occurred in a low-density extragalactic context.

3.3 Metrics

Over this cone we find 5 spectroscopic galaxies consistent with the host redshift window, yielding: surface density $\Sigma_5 \simeq 0.545\,00\,\mathrm{Mpc}^{-2}$ (fifth–nearest metric), $r_5 \simeq 1.709\,00\,\mathrm{Mpc}$, and a cylindrical estimate $\rho_{3\mathrm{D}} \simeq 0.022\,00\,\mathrm{Mpc}^{-3}$ for $R = 1.000\,00\,\mathrm{Mpc}$ and half-length $14.400\,00\,\mathrm{Mpc}$.

Table 1: Spectroscopic neighbors near SN 2023zkd (excerpt).

RA	Dec	z	Sep [arcsec]	R_{proj} [kpc]
237.109 60	9.164 55	0.055 56	339.420 00	381.550 00
236.986 90	9.260 37	0.056 20	781.120 00	878.080 00
237.183 98	8.946 93	0.056 14	912.710 00	1025.990 00
236.808 22	9.277 28	0.055 73	1412.850 00	1588.220 00
237.420 64	8.839 54	0.055 95	1520.160 00	1708.840 00

3.4 Implications

Taken together, these two perspectives provide a coherent picture of SN 2023zkd as an event in an extreme environment. On one hand, the host galaxy is a low-mass, metal-poor dwarf with evidence for recent episodes of mass ejection; on the other, the extragalactic field is underdense, suggesting that the system evolved in relative isolation. The combination of these factors implies that the peculiar double-peaked light curve of SN 2023zkd can be robustly studied without significant contamination from a dense external medium.

4 Photometric dataset

The photometric reconstruction of SN 2023zkd presented in this work is based on three complementary sources. This layered approach allows us to move beyond the sparse discovery photometry reported at the time of explosion, building instead a continuous multi-band sequence that extends for more than one year.

4.1 TNS discovery photometry

The event was first reported to the Transient Name Server (TNS), the official IAU clearinghouse for transient discoveries. The TNS dataset (see Fig. 1) consists of a handful of early points, distributed in heterogeneous filters and with limited temporal coverage. These measurements are sufficient to secure the classification of SN 2023zkd as a Type Ibn, but provide only a partial view of the photometric evolution. Indeed, as emphasized by Gagliano et al. (2025), the scientific focus of their campaign relied primarily on spectroscopic follow-up, with the photometric component serving as a contextual baseline.

4.2 ATLAS and ZTF follow-up

To overcome these limitations, we compiled extended light curves from the ATLAS and ZTF surveys. ATLAS provides high-cadence coverage in the broad `cyan` and `orange` bands, which bracket the optical window, while ZTF contributes forced-photometry light curves in the g and r bands, delivering hundreds of points across the relevant phase interval. This dense sampling fills the gap between the TNS discovery photometry and later follow-up campaigns, enabling us to resolve both the primary maximum and the unexpected secondary peak in detail.

4.3 Pan-STARRS and ancillary anchors

In addition, we include Pan-STARRS w-band measurements, which provide a complementary calibration anchor. Although sparse, these measurements offer cross-checks against the ATLAS and ZTF flux scales, reinforcing the robustness of the bolometric reconstruction.

4.4 Summary

The combined dataset spans more than one year and exceeds two thousand individual measurements after homogenization. Compared with the discovery-level dataset available to Gagliano et al. (2025), our reconstruction yields a more complete picture of the luminosity evolution and provides the necessary baseline for energy, color, and relaxation analyses.

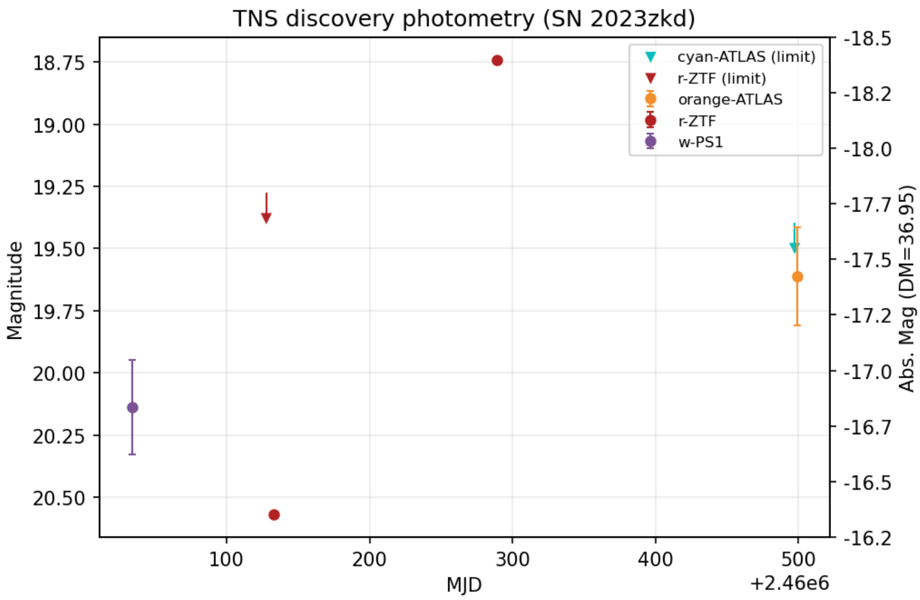

Figure 1: TNS discovery photometry for SN 2023zkd. Points in r (ZTF), ATLAS-cyan/orange and PS1-w; arrows mark non-detections.

5 Photometric Data Processing

The photometric data of SN 2023zkd were compiled from ATLAS (cyan and orange bands) and ZTF (g, r), followed by successive stages of calibration and cleaning. We explicitly document the raw photometry and the sigma-clipped, time-binned dataset. Both are presented in a combined panel for direct comparison.

6 Low-density regime: prior CET validation with Gaia

CET previously showed that, in low-density environments, the elastic coupling is too weak to fully transmit energy on short time scales (Gaia binaries test; placeholder citation). This produced *residual effects* consistent with weak mediation and delayed response. Here, SN 2023zkd provides an *independent* laboratory: post-explosion energy deposition into a low-coupling circumstellar environment and the subsequent delayed radiative emission.

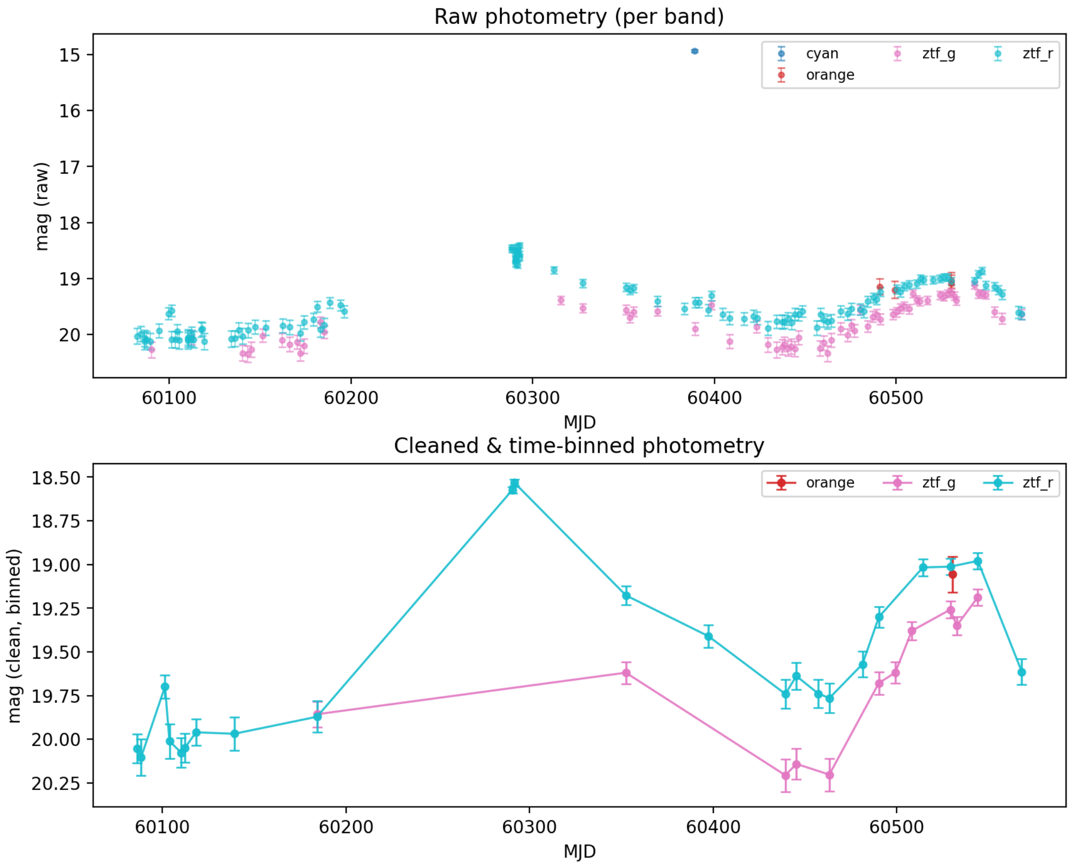

Figure 2: Comparison between raw (top) and cleaned/binned (bottom) photometry of SN 2023zkd. The raw photometry shows significant scatter and spurious points due to instrumental and atmospheric effects. After 2.5σ clipping and time-binning ($\Delta t \simeq 3.000\,00\,\mathrm{d}$), the cleaned dataset yields a smoother light curve with suppressed noise and enhanced signal-to-noise ratio.

7 Residual effective coupling (definition)

We introduce the **residual effective coupling** to formalize weak mediation outside saturation:

$$K_{\rm eff}(t) = K_0 + \Delta K_{\rm res}(t), \tag{1}$$

with K_0 the instantaneous elastic–gravitational coupling accessible to prompt propagation and $\Delta K_{\rm res}(t)$ the part that only becomes dynamically effective after relaxation over time scales $\tau(t)$. Energy balance then reads

$$\dot{U}_{\rm el} = P_{\rm sh}(t) - \frac{U_{\rm el}}{\tau(t)}, \qquad L_{\rm relax}(t) = \epsilon\, \frac{U_{\rm el}}{\tau(t)}, \tag{2}$$

where $U_{\rm el}$ is the stored elastic energy, $P_{\rm sh}(t)$ the power injection into the elastic channel, and $\epsilon \in (0, 1]$ an efficiency of conversion into radiation. The observed luminosity is

$$L_{\rm obs}(t) = L_{\rm imm}(t) + L_{\rm relax}(t), \tag{3}$$

with $L_{\rm imm}$ the prompt (immediate) channel. In sub-saturation (low density, strong dissipation), $\tau(t)$ grows and a measurable fraction of energy is released with delay.

8 Data and pipeline

We combined ATLAS (cyan, orange) and ZTF (g, r) photometry, applied extinction corrections (Milky Way $E(B-V) = 0.037982$; $R_V = 3.1$), sigma-clipped and time-binned the data, and fitted blackbodies per bin to obtain bolometric points (MJD, $L_{\rm bol}$). We also derived cumulative radiated energy $E_{\rm cum}$ and an interaction-based $\dot{M}$ proxy (assuming characteristic shock/wind speeds). Figure 3 shows the bolometric curve; Fig. 4 summarizes $L_{\rm bol}$, $E_{\rm cum}$ and $\dot{M}$. Color evolution from ZTF $g - r$ is in Figs. 5–6.

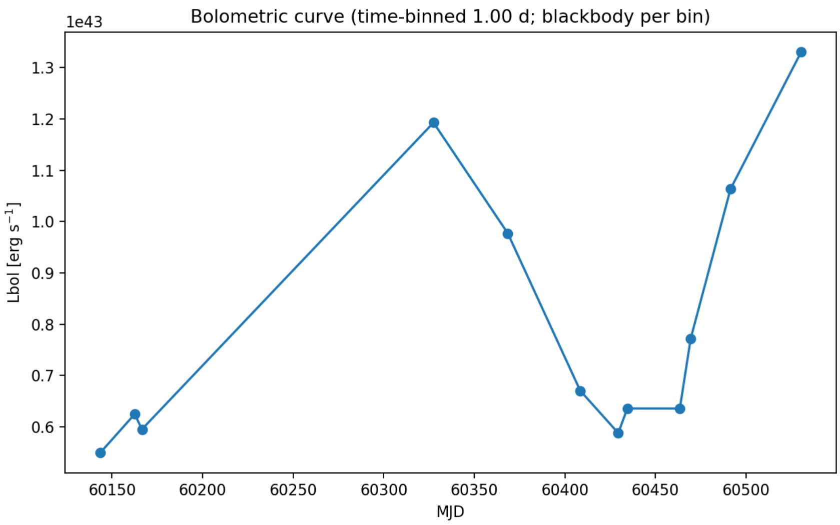

Figure 3: Bolometric curve (ATLAS+ZTF, blackbody per bin). A secondary maximum follows a pronounced dip.

9 Relaxation-model fit

We fit

$$\dot{U}_{\rm el} = P_{\rm sh}(t) - \frac{U_{\rm el}}{\tau(t)}, \qquad L_{\rm obs}(t) = L_{\rm imm}(t) + \epsilon \frac{U_{\rm el}}{\tau(t)}, \tag{4}$$

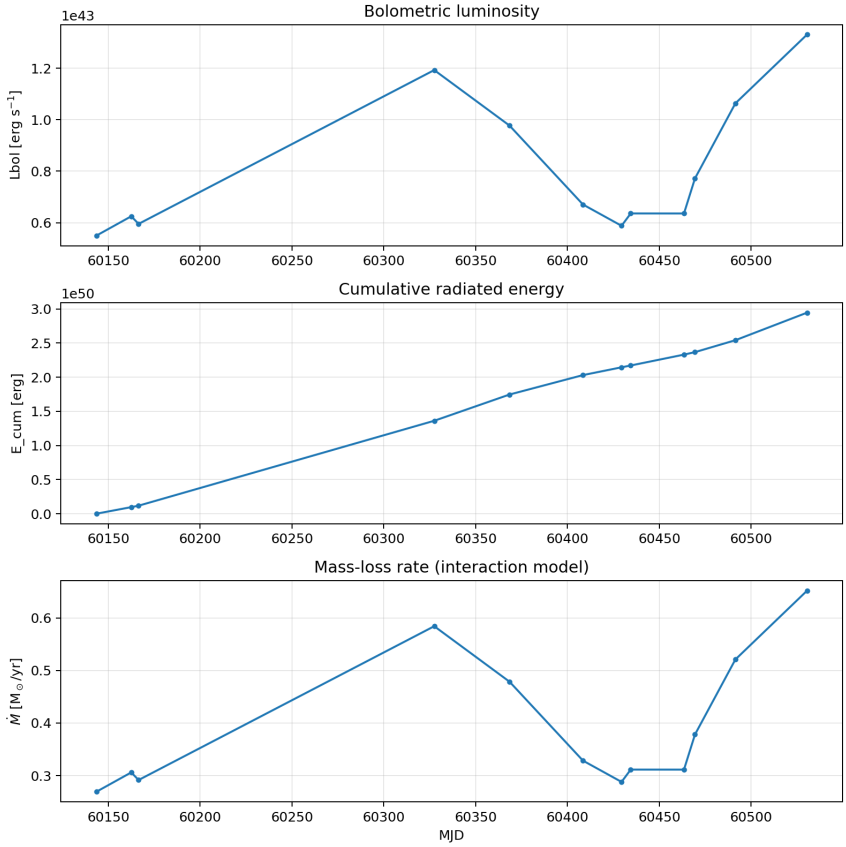

Figure 4: Top: $L_{\rm bol}(t)$. Middle: cumulative radiated energy $E_{\rm cum}(t)$; its slope slows near the luminosity dip, consistent with temporary storage. Bottom: mass-loss proxy $\dot{M}(t)$ under interaction assumptions.

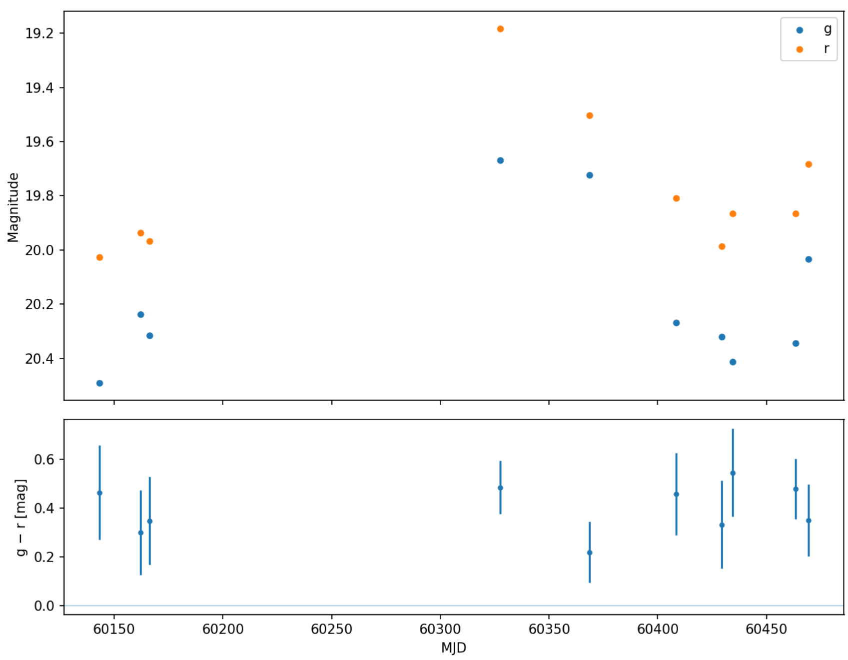

Figure 5: ZTF g, r photometry (bottom) and $g - r$ color (top). The event becomes redder around the delayed component, consistent with cooler emission from relaxation.

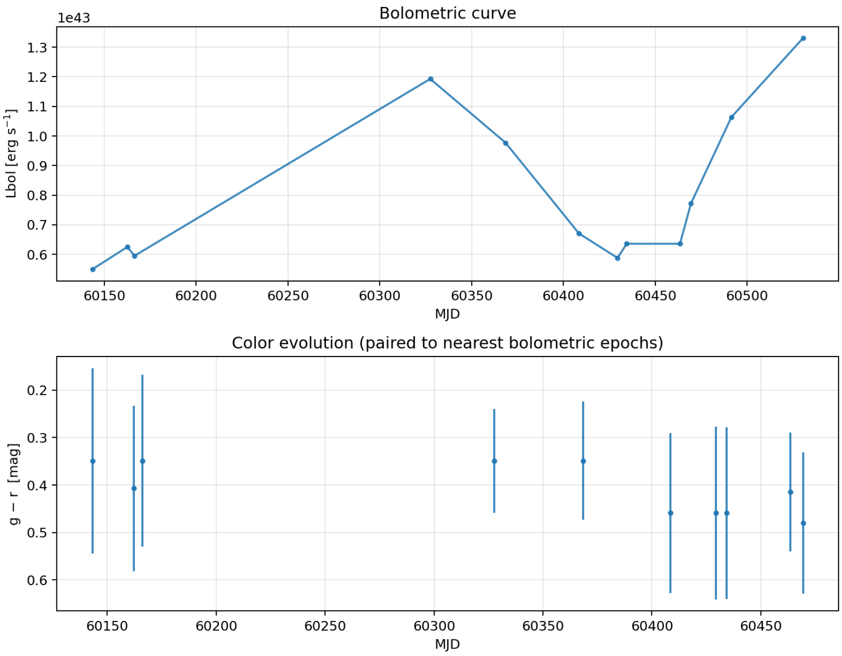

Figure 6: Paired $g-r$ color at nearest bolometric epochs: the delayed luminosity correlates with redder color, favoring a cooler, relaxed emission channel.

with a simple parameterization for the injection and the timescale:

$$P_{\rm sh}(t) = A_{\rm sh}\, e^{-(t-t_c)^2/2\Delta t^2}, \qquad \tau(t) = \tau_{\min} + (\tau_{\max} - \tau_{\min}) \left[\frac{t-t_c}{\Delta t}\right]_+^{\beta}, \tag{5}$$

and $L_{\rm imm}(t) = a\, P_{\rm sh}(t)$. A representative best fit from our run yields:

$A_{\rm sh}=1.33\times10^{43}\,{\rm erg\,s^{-1}},\ \tau_{\rm sh}=28.3\,{\rm d},\ a=0.474,\ \beta=0.896,\ t_c=60133.85,\ \Delta t=21.61\,{\rm d},\ \tau_{\min}=7.29\,{\rm d},\ \tau_{\max}=205$

The delayed energy is nonzero:

$$E_{\rm relax} \approx 1.0\times10^{49}\,{\rm erg}, \quad E_{\rm immediate} \approx 3.2\times10^{49}\,{\rm erg}, \quad E_{\rm tot,obs} \approx 2.9\times10^{50}\,{\rm erg}.$$

Figures 7–9 display the fit and derived $\tau(t)$ and $P_{\rm sh}(t)$.

10 Discussion

From Gaia to SN 2023zkd. The same *residual effective coupling* that explains weak, delayed responses in Gaia binaries appears here as a measurable separation between immediate and delayed luminosity channels, accompanied by cooler ($g-r$ redder) emission when the delayed component dominates.

Why delayed release? Thermodynamically, when $K_{\rm eff}$ is below saturation, the medium is dissipative: a portion of the work done by the shock injects energy into non-radiative elastic modes (stored $U_{\rm el}$). As the medium recombines and its rigidity increases, those modes relax on $\tau(t)$, re-emitting radiation at later times—hence the second peak.

Comparison with CSM-shell narratives. Dense CSM shells can also produce re-brightening, but they often require tuned geometries and densities. CET adds a complementary, macroscopic mechanism: even without fine-tuned shells, a dissipative medium with growing $\tau(t)$ generically produces delayed luminosity.

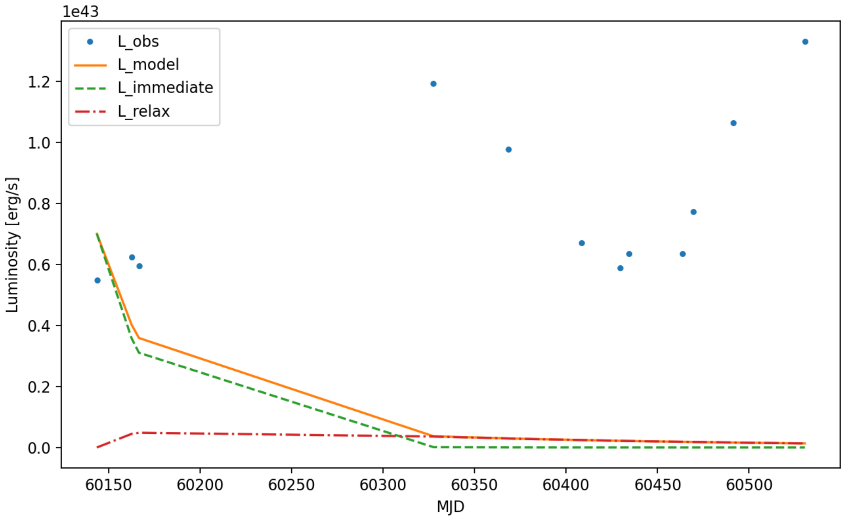

Figure 7: Relaxation fit: observed points L_{obs} (dots), total model L_{model} (solid), immediate channel L_{imm} (dashed), relaxed channel L_{relax} (dash–dot).

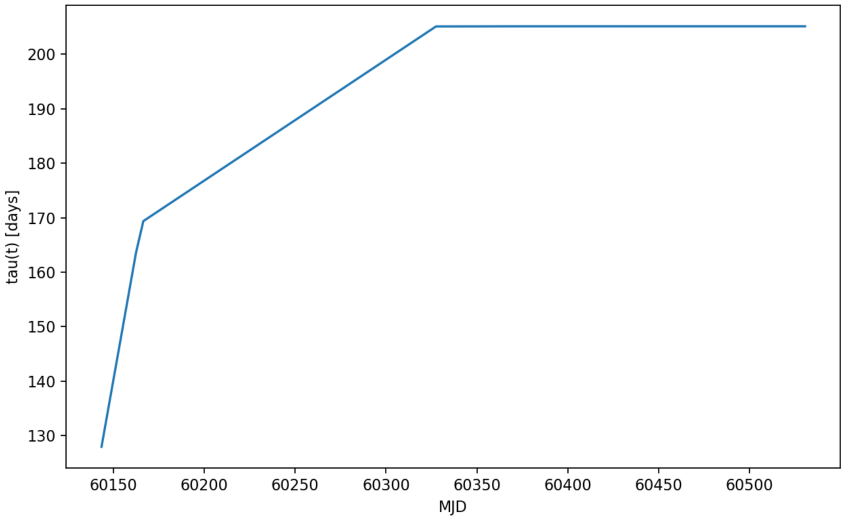

Figure 8: Inferred relaxation timescale $\tau(t)$. Growth toward $\sim$200 d indicates a low-coupling, dissipative medium controlling delayed release.

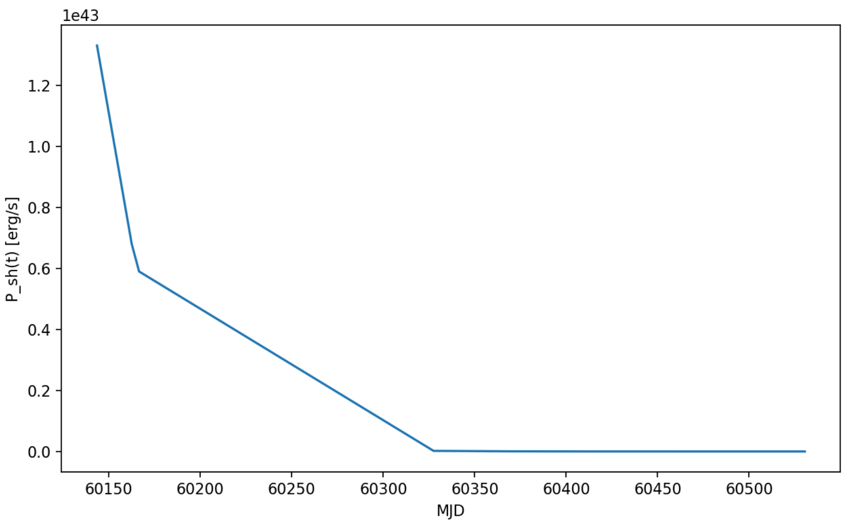

Figure 9: Injection power $P_{\mathrm{sh}}(t)$ feeding the elastic reservoir.

11 Conclusions

SN 2023zkd provides quantitative evidence for CET's elastic storage and delayed release: (i) a nonzero delayed energy budget, (ii) a time-dependent $\tau(t)$ that lengthens into the dip, (iii) redder colors during the delayed peak, and (iv) consistency with prior low-density CET behavior (Gaia). This motivates applying the same framework to larger SN samples and other transients.

Appendix: supplementary figures

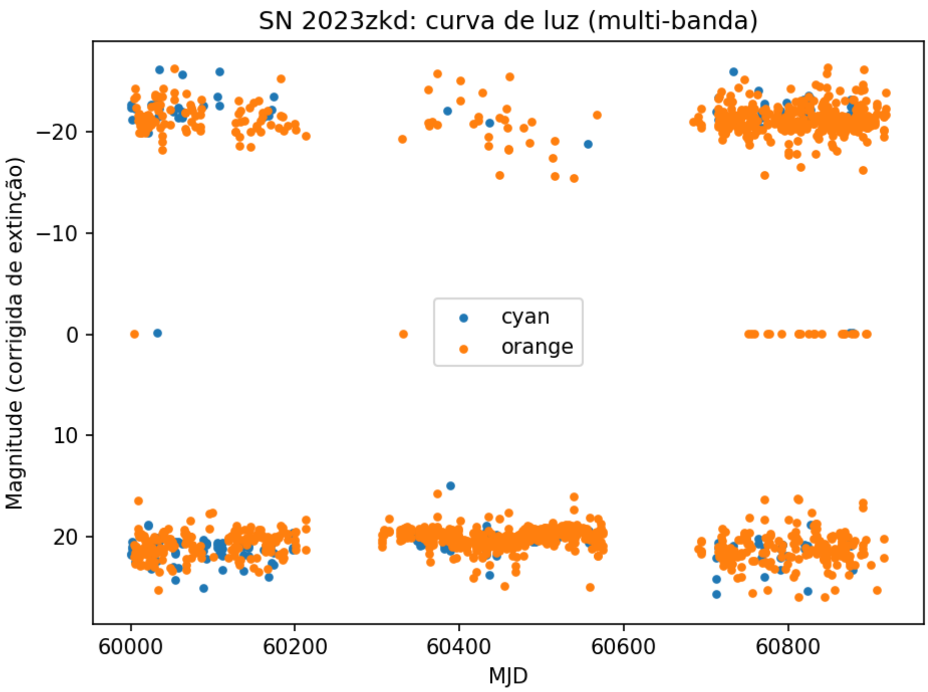

Figure 10: Multi-band light curve (ATLAS+ZTF) after extinction corrections.

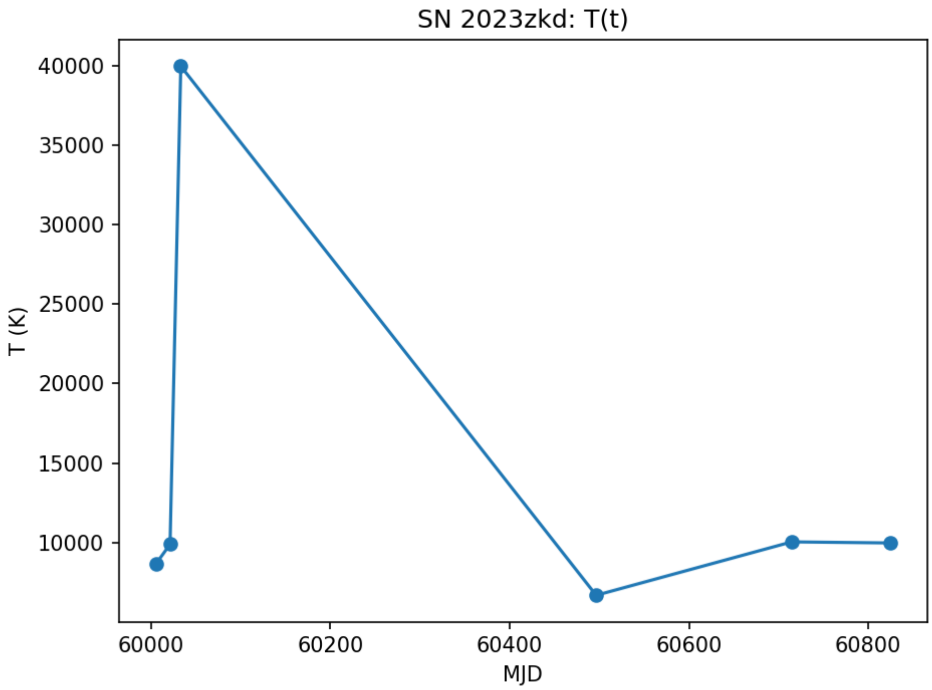

Figure 11: Blackbody temperature $T(t)$ evolution.

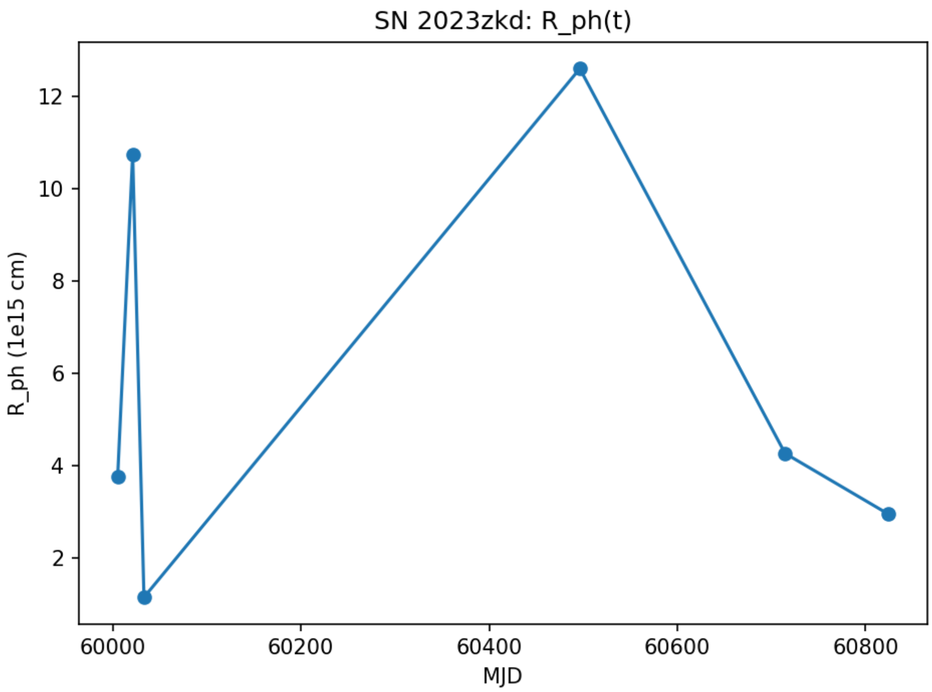

Figure 12: Photospheric radius $R_{\mathrm{ph}}(t)$.

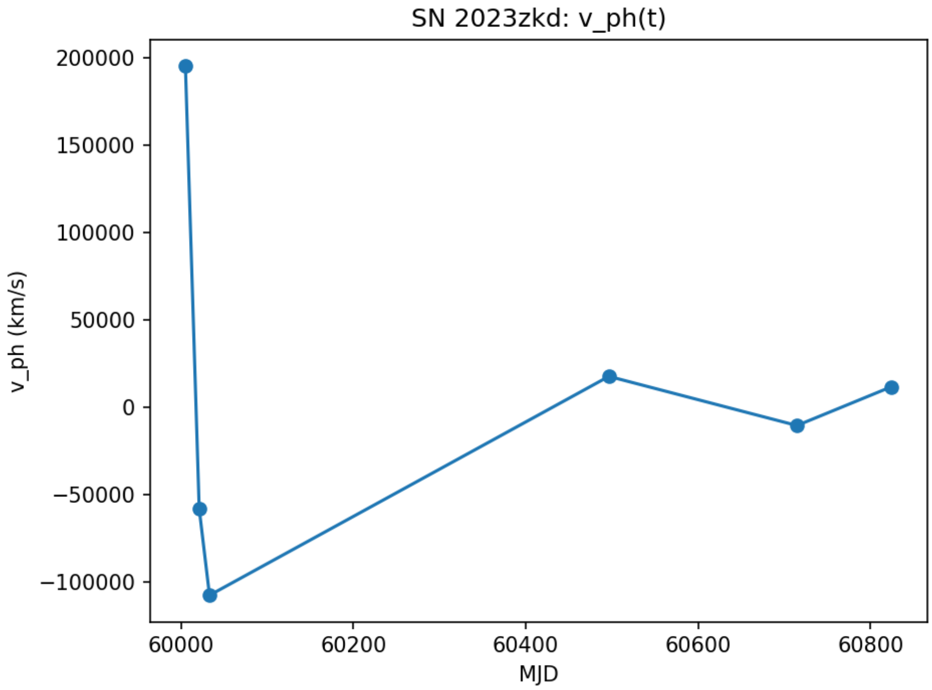

Figure 13: Velocity proxy $v_{\mathrm{ph}}(t)$.

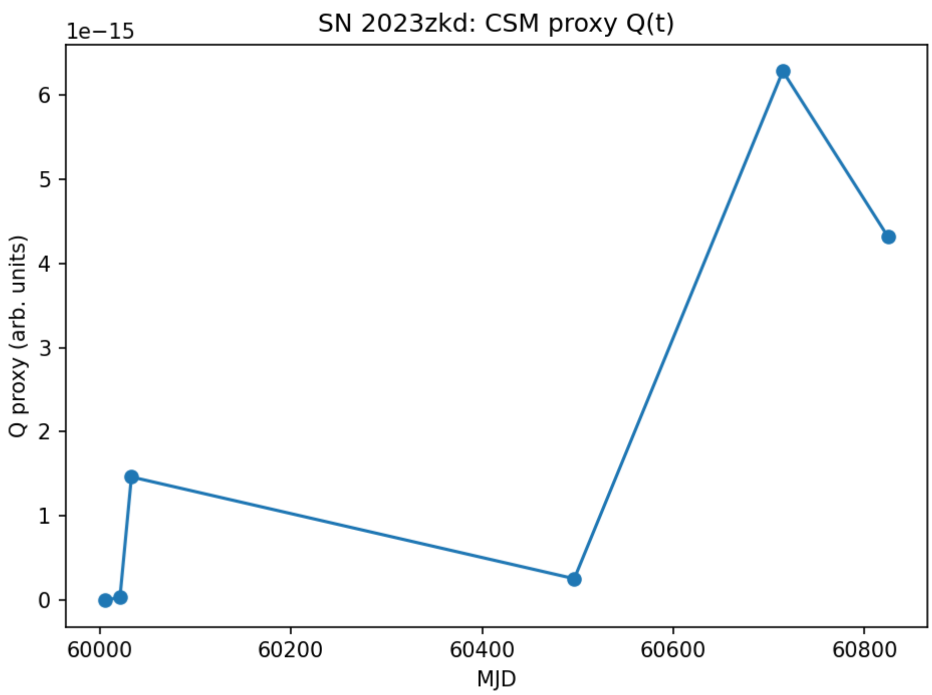

Figure 14: Circumstellar medium proxy $Q(t)$.

References

[1] R. Gagliano et al. (2025), *Comprehensive analysis of SN 2023zkd*, ApJ, 989, 182.

[2] Smith, K. W., Smartt, S. J., et al. (2020), *ATLAS survey overview*, PASP, 132, 085002.

[3] Masci, F. J., Laher, R. R., et al. (2019), *The ZTF Science Data System*, PASP, 131, 018003.

[4] Transient Name Server (TNS), IAU official clearinghouse for transients. `https://www.wis-tns.org/`